

Chakra

See where your carbon leaks. Close the loop with numbers that pay.

Morbi Tile Works

Ceramic tiles and sanitaryware · Gujarat · assessed 12 September 2026

ANNUAL FOOTPRINT

26,075

tCO₂e · range 24,183–28,213
(± 7.7%)

LARGEST LEAK POINT

67.1%

Process heat (boiler / furnace fuel)

PAYS FOR ITSELF

₹9.08 Cr

per year · 14 interventions · 13 mo payback

ABATEMENT AT NO NET COST

28.4%

7,393 tCO₂e · 30.1% available in total

This plant emits about 26,075 tCO₂e a year. The largest single leak point is process heat (boiler / furnace fuel) at 67 percent of the total. 14 of the recommended interventions pay for themselves: Rs 908.0 lakh a year of net benefit for Rs 988.1 lakh of capital, removing 28 percent of the footprint.

Emission inventory

Source	Scope	Activity	tCO ₂ e	Share	Factor source
Process heat (boiler / furnace fuel)	Scope 1	298,710 GJ/yr	17,488	67.1%	IPCC / DEFRA combustion factors
Purchased electricity	Scope 2	8,400,000 kWh/yr	5,880	22.6%	CEA regional grid mix - Gujarat
Inbound and outbound freight	Scope 3	21,000,000 tonne-km/yr	1,995	7.7%	DEFRA modal freight factors
Purchased Virgin kraft / corrugate	Scope 3	520 tonne/yr	650	2.5%	CEPI / DEFRA material factors
Inert waste to landfill	Scope 3	3,100 tonne/yr	62	0.2%	DEFRA waste disposal

Scope 1 17,488 · Scope 2 5,880 · Scope 3 2,707 tCO₂e. Biogenic CO₂ of 0 t is reported separately and excluded from the Scope 1 total per GHG Protocol.

Leak points

Process heat (boiler / furnace fuel)

CRITICAL

· 67.1% of footprint · 17,488 tCO₂e

This stream is 67 percent of the total footprint and runs above the sector median (7 vs 6). It is not an outlier, but it is large enough that a median-level improvement is worth about 3,964 tCO₂e a year.

Purchased electricity

MODERATE

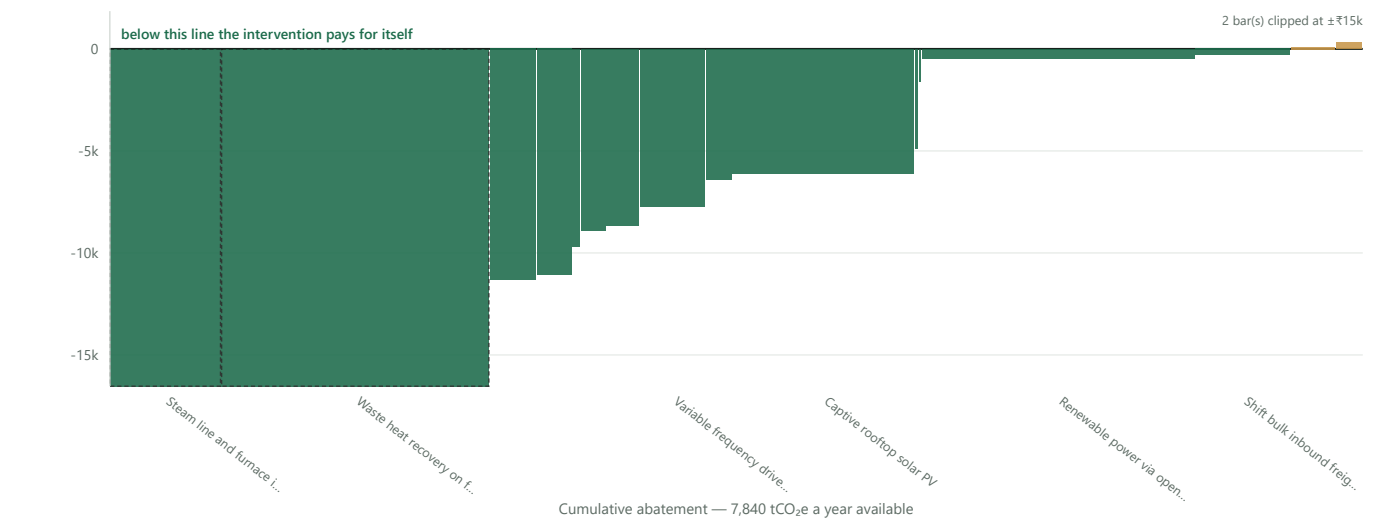
· 22.6% of footprint · 5,880 tCO₂e

This stream is 23 percent of the total footprint and runs above the sector median (200 vs 195). It is not an outlier, but it is large enough that a median-level improvement is worth about 147 tCO₂e a year.

Benchmarks are indicative screening percentiles compiled for this build from published sector energy-intensity literature and cluster study ranges. They are good enough to rank a plant against its peers and to flag outliers, and are NOT a substitute for a BEE-accredited audit. Replace with cluster-specific survey data before production use.

Benchmarks are literature priors; no corpus data yet for this sector.

Marginal abatement cost curve



Bar width is tCO₂e abated a year; height is rupees per tonne. Green bars have a negative cost of abatement — they pay for themselves. Bars are de-rated for interaction: each intervention applies to what the one before it left behind.

Costed interventions

Intervention	Type	tCO ₂ e/yr	₹/tCO ₂ e	Capex	Net ₹/yr	Payback
Steam line and furnace insulation upgrade Process heat (boiler / furnace fuel) · high confidence · difficulty 1/5 · 2d downtime	process	700	-21,509	₹15.8 L	₹1.54 Cr	1 mo
Waste heat recovery on flue gas Process heat (boiler / furnace fuel) · high confidence · difficulty 3/5 · 7d downtime	process	1,679	-20,639	₹1.47 Cr	₹3.85 Cr	5 mo
Idle load elimination and production scheduling Purchased electricity · medium confidence · difficulty 1/5 · 0d downtime	process	294	-11,327	₹1.1 L	₹33.6 L	immediate
Compressed air leak survey and repair Purchased electricity · high confidence · difficulty 1/5 · 0d downtime	process	223	-11,083	₹2.0 L	₹26.6 L	1 mo
Power factor correction and capacitor bank Purchased electricity · high confidence · difficulty 2/5 · 1d downtime	energy	54	-9,666	₹4.7 L	₹6.5 L	9 mo
LED lighting retrofit Purchased electricity · high confidence · difficulty 1/5 · 1d downtime	energy	159	-8,886	₹22.7 L	₹20.2 L	13 mo
Recycled content corrugate and packaging CAPPED Purchased Virgin kraft / corrugate · high confidence · difficulty 1/5 · 2d downtime	material	211	-8,642	₹1.9 L	₹18.7 L	1 mo
Variable frequency drives on pumps and fans Purchased electricity · high confidence · difficulty 2/5 · 3d downtime	energy	412	-7,707	₹1.08 Cr	₹53.6 L	2.0 yr
Premium efficiency motor replacement Purchased electricity · high confidence · difficulty 2/5 · 2d downtime	energy	166	-6,395	₹70.6 L	₹23.5 L	3.0 yr
Captive rooftop solar PV Purchased electricity · high confidence · difficulty 3/5 · 0d downtime	energy	1,143	-6,119	₹6.09 Cr	₹1.68 Cr	3.7 yr
Sell process residue as another industry feedstock All waste streams · medium confidence · difficulty 2/5 · 3d downtime	waste	22	-4,898	₹1.2 L	₹1.4 L	10 mo
Source segregation and recycling contract All waste streams · high confidence · difficulty 1/5 · 2d downtime	waste	18	-1,615	₹1.6 L	₹90.0k	22 mo
Renewable power via open access or green tariff Purchased electricity · medium confidence · difficulty 3/5 · 0d downtime	energy	1,714	-484	₹2.7 L	₹14.7 L	2 mo
Shift bulk inbound freight from road to rail Inbound and outbound freight · medium confidence · difficulty 3/5 · 0d downtime	logistics	598	-297	₹81.0k	₹2.0 L	5 mo
Supplier radius reduction Inbound and outbound freight · low confidence · difficulty 3/5 · 0d downtime	logistics	279	94	₹1.4 L	₹0	—

Intervention	Type	tCO ₂ e/yr	₹/tCO ₂ e	Capex	Net ₹/yr	Payback
Outbound load consolidation and route planning Inbound and outbound freight · medium confidence · difficulty 2/5 · 0d downtime	logistics	168	321	₹2.0 L	-₹40.0k	—

Start here: 8 quick wins — payback under two years and low implementation difficulty — need ₹51.0 L of capital and return ₹2.62 Cr a year.

Considered and rejected

✗ **Fuel switch to agri-residue briquette**
Blocked: tunnel kiln firing requires stable high-temperature flame characteristics and low ash carryover onto glazed surfaces. Briquette firing is not qualified for glazed tile kilns. Spray dryer duty may be convertible and is assessed separately.

Compliance exposure

CBAM

INDICATIVE ANNUAL EXPOSURE

₹0

0 tCO₂e embedded at 0.0% EU export share

Basis. Scope 1 plus Scope 2 embedded emissions apportioned by EU export share. Precursor and upstream material emissions are excluded - CBAM precursor rules require supplier-specific data this tool does not hold, and estimating them would be a guess presented as a number.

Indicative exposure only. The CBAM definitive regime requires installation-level verified data and specific CN code mapping. Confirm your product CN codes and the current certificate price before acting on this figure.

BRSR readiness

Scope 1 inventory	ready	17488.0
Scope 2 inventory (location-based)	ready	5880.0
Scope 3 - purchased goods, waste, transport	partial	2707.0
Emission intensity per unit output	ready	
Third-party assurance	not started	

Methodology and limitations

GHG Protocol Corporate Accounting and Reporting Standard · IPCC AR6 GWP100 · grid factor: CEA regional grid mix - Gujarat.

Factors. Every factor carries a source and an uncertainty band. The engine propagates the band to every result. Point estimates are never shown without their range.

Verification status. HACKATHON BUILD - factors are literature values and must be re-verified against the primary source editions cited before commercial use.

Leak rule. A stream is flagged as a LEAK POINT when its intensity exceeds the sector p75, or when it contributes more than 15 percent of total footprint AND sits above p50. Severity scales with the distance above p75.

Interaction. Portfolio totals apply each intervention to the RESIDUAL stream in cost order, not to the original baseline. Standalone and de-rated abatement are both shown on every card. Cross-stream overlap (for example in-house regrind against a material substitution) is not de-rated and is declared as a known limitation.

Capex. Screening-grade intervention library. Every economic figure is an order-of-magnitude planning estimate intended to rank options, not to replace a vendor quotation. The UI states this on every card.

Screening assessment — not a BEE-accredited energy audit and not an assurance engagement. Economics are planning-grade estimates intended to rank options and justify obtaining a vendor quotation, not to replace one. Assumptions: electricity at ₹8.0/kWh, cost of capital 12%.